

FAIRPORT, MI, USA

WMO: 997792

Lat: 45.6187N

Lon: 86.6596W

Elev: 186

StdP: 99.11

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.2	(c) -17.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 15.2	(k) -9.5	(l) 13.8	(m) -7.9	(n) 6.2	(o) 300	(p) 0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.6	(c) 24.7	(d) N/A	(e) 23.7	(f) N/A	(g) 22.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.6	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.9	(b) 11.6	(c) 10.5	(d) DB	(e) -22.1	(f) 27.1	(g) 4.3	(h) 1.8	(i) -25.2	(j) 28.4	(k) -27.8	(l) 29.5	(m) -30.2	(n) 30.5	(o) -33.4	(p) 31.9	(q) N/A
(a) 12.9	(b) 11.6	(c) 10.5	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-7.4	-7.4	-2.4	3.5	9.4	14.8	18.7	18.9	15.4	9.4	3.1	-3.0
	DBStd	10.14	5.74	5.49	5.20	3.29	3.15	3.00	2.85	2.36	3.28	3.69	4.42	4.60
	HDD10.0	2325	539	487	384	194	49	2	0	0	2	56	208	402
	HDD18.3	4536	797	720	643	444	278	112	31	21	97	276	457	661
	CDD10.0	925	0	0	0	1	29	145	269	277	163	39	1	0
	CDD18.3	93	0	0	0	0	0	4	42	39	8	0	0	0
	CDH23.3	118	0	0	0	0	1	4	69	42	2	0	0	0
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0
Wind	WSAvg	4.7	5.4	4.8	4.4	4.6	4.3	3.7	3.9	4.2	4.7	5.4	6.1	5.4
Precipitation	PrecAvg	756	45	32	48	60	76	71	72	88	86	66	57	55
	PrecMax	986	119	102	151	102	188	182	142	168	180	134	141	135
	PrecMin	547	4	1	3	14	22	9	17	5	22	20	12	3
	PrecStd	116	27	23	35	22	36	40	31	41	39	30	32	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.9	3.7	10.9	13.2	20.6	24.1	26.8	25.6	22.9	18.8	12.6	8.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	2.8	1.9	7.5	11.0	17.6	21.7	25.1	24.5	21.7	17.3	11.2	5.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	1.8	1.1	5.8	9.5	15.5	20.2	24.0	23.6	20.9	16.3	10.1	4.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	0.7	0.3	3.8	8.1	13.8	19.1	22.9	22.7	20.2	15.2	9.1	3.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.0	6.7	5.8	6.5	6.0	5.6	5.6	5.6	5.0	4.7	5.1
		MCDBR	5.2	5.7	6.4	7.9	8.6	7.1	6.1	5.9	5.7	4.9	4.4	5.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.269	0.282	0.313	0.355	0.386	0.393	0.389	0.387	0.369	0.335	0.303	0.274	
	taud	2.492	2.416	2.401	2.373	2.355	2.386	2.412	2.412	2.457	2.523	2.536	2.516	
	Ebn at Noon	859	908	920	904	882	874	873	862	849	835	807	813	
	Edh at Noon	65	85	101	114	120	117	113	109	96	78	64	58	
All-Sky Solar Radiation	RadAvg	1.32	2.29	3.55	4.60	5.62	6.05	6.22	5.26	4.00	2.42	1.37	1.02	
	RadStd	0.12	0.27	0.33	0.43	0.44	0.48	0.26	0.35	0.29	0.26	0.14	0.09	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page